

Crypto Data Science Assignment Report

Prepared for the Hyperliquid / Fear-and-Greed trading analysis assignment.

1. Task Summary

The assignment asks for an analysis of the relationship between trader behavior on Hyperliquid and Bitcoin market sentiment, using a historical execution dataset and the daily Fear and Greed Index. The core goal is to identify how sentiment regimes affect performance, execution style, and risk, and then use those findings to build a leakage-safe predictive model.

2. Data Overview

Historical Trader Data

- 211,224 rows, 32 unique accounts, 246 unique coins.
- Main fields: account, coin, execution price, size, side, timestamp IST, start position, direction, closed PnL, crossed, fee, trade ID.
- Data is heavy-tailed, with large outliers in `Closed PnL` and `Size USD`.

Fear and Greed Index

- 2,644 daily rows spanning 2018-02-01 to 2025-05-02.
- Fields: timestamp, value, classification, date.
- Classification buckets: Extreme Fear, Fear, Neutral, Greed, Extreme Greed.

3. Preprocessing and Join Logic

The most important technical step is the timestamp alignment. The trade data is in IST, while sentiment is daily and indexed by date. To avoid leakage and off-by-one-day mismatches, the pipeline converts trade timestamps from IST to UTC and then joins on the UTC date key.

1. Parse `Timestamp IST` as `DD-MM-YYYY HH:MM`.
2. Subtract 5 hours 30 minutes to obtain UTC time.
3. Derive `date_utc` and join to the sentiment `date` field.
4. Winsorize `Closed PnL` and `Size USD` at the 1st and 99th percentiles for robust statistics.
5. Remove spot dust conversion rows and similar administrative noise from analysis tables.

4. Exploratory Findings

- Performance differs materially across sentiment regimes.

- Maker/taker behavior shifts with fear and greed; traders become more aggressive in greed regimes.
- Trade sizing and leverage proxies increase in higher sentiment regimes, indicating stronger risk-taking.
- Account performance is heterogeneous: a few accounts dominate total realized PnL.

Regime-Level EDA Outputs

- `outputs/eda_sentiment_regimes.png`
- `outputs/quant_regime_core_metrics.png`
- `outputs/quant_regime_transition_matrix.png`
- `outputs/quant_coin_microstructure.png`

5. Advanced Quant Analysis

The advanced notebook extends the analysis beyond basic EDA and includes account-level risk decomposition, behavioral clustering, regime specialization, coin-level microstructure, and bootstrap confidence intervals for regime metrics.

- Account skill map: win rate vs average PnL.
- Risk map: max drawdown vs total realized PnL.
- KMeans behavioral clustering for trader archetypes.
- Regime specialization heatmap by account.

6. Predictive Modeling Pipeline

The modeling notebooks build a leakage-safe classifier for $C_{\text{losed}} \text{ PnL} > 0$ using lagged and rolling features only. The validation strategy is walk-forward with purging and embargo. The champion model is a boosted-tree baseline combined with thresholding, confidence-based sizing, and a final guard gate.

- Primary metric set: ROC-AUC, PR-AUC, F1, Brier score.
- Trading metric set: total PnL, max drawdown, hit rate, profit factor.
- Thresholds and confidence sizing are optimized out-of-sample on walk-forward predictions.

Model Outputs

- `csv_files/model_fold_scores.csv`
- `csv_files/model_oof_predictions.csv`
- `csv_files/final_champion_summary.csv`
- `model_artifacts/final_calibrated_hgb_model.joblib`
- `model_artifacts/final_model_metadata.json`
- `outputs/final_champion_equity.png`

- `outputs/final_champion_drawdown.png`
- `outputs/final_champion_regime_pnl.png`

7. Final Champion Result

Item	Value
Model	baseline_hgb_final_champion
Best threshold	0.62
Best alpha	1.5
Best confidence gate	0.05
Taken trades	143,880
Total PnL	14,599,108
Max drawdown	-89,678.22
Hit rate	95.02%
Profit factor	40.22

Relative to the earlier baseline champion, the final pass materially improved total PnL while keeping drawdown close to the same risk envelope. The strongest gain came from confidence-based sizing and the safety gate that removed middle-confidence trades.

8. Untouched Final Holdout

Item	Value
Model	final_holdout_calibrated_hgb
Best threshold	0.52
Best alpha	0.75
Best	0.05

confidence
gate

Holdout rows	31,663
Taken trades	11,752
Total PnL	883,481.89
Max drawdown	-58,738.89
Hit rate	93.13%
Profit factor	11.28

This holdout was never used for threshold or sizing selection. It provides a cleaner estimate of generalization than the walk-forward OOF champion alone and still shows profitable, controlled behavior.

9. Explainability And Cost Stress

Permutation importance on the untouched holdout shows that the strongest driver is trade direction, followed by recent account win rate and account PnL momentum. That is a very plausible structure for a market-aware trading model: execution intent and trader state matter more than raw sentiment alone.

Top Holdout Features	Relative Importance
Direction	0.277188
acc_wr20	0.039854
acc_pnl20	0.007524
Side	0.002160
Start Position	0.001739

Cost stress is also favorable: the holdout remains profitable at 5 bps and 10 bps of extra execution cost, which suggests the signal is not purely a zero-cost artifact.

The final modeling design also uses the strongest practical ideas from the advanced modeling notes: calibrated probabilities for position sizing, lagged sentiment velocity, account momentum features, and purged time-series evaluation. Triple-barrier labeling and full graph-based wash-

trading loops are documented as future extensions because the available fill-level dataset does not contain a clean future path or wallet-transfer graph.

10. What Is Covered vs the Guidance PDF

- Covered: sentiment join, regime analysis, maker/taker behavior, clustering, leakage-safe modeling, purged walk-forward validation, copy-trading style risk controls.
- Covered: final champion selection with constrained grid search and OOS backtest metrics.
- Partially covered: literal wash-trading graph heuristics and inverse covariance clustering as named in the guidance.
- Missing as a formal file: a finalized PDF report artifact and a fully mirrored raw-data folder structure inside `csv_files/`.

11. Recommended Submission Bundle

1. `notebook_1.ipynb` and `notebook_1_advanced_quant.ipynb`
2. `notebook_2.ipynb` and `notebook_2_improved.ipynb`
3. `ds_report.pdf`
4. `csv_files/` with raw and processed data
5. `outputs/` with the major charts

12. Conclusion

The assignment is substantially completed with a strong, leakage-aware quant workflow, advanced regime analysis, an untouched holdout evaluation, and a production-style champion model. The remaining work is mostly packaging and presentation polish rather than core analytical gaps.